

Practical 9

system has RAM which can store four pages simultaneously to satisfy page

requests. Write a program to calculate percentage page hit and page fault if the

system uses Optimal page replacement technique for the page references

entered by the user i.e. 1, 2,3,4, L,2,5, !,2,3,4, 5 (No of Frames=3)'

```
tanuja@Tanuja: ~  
GNU nano 6.2  
#include <stdio.h>  
  
int main()  
{  
    int frames = 3;  
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int frame[3];  
    int i,j,k;  
  
    int pageFault = 0, pageHit = 0;  
  
    for(i=0;i<frames;i++)  
        frame[i] = -1;  
  
    for(i=0;i<12;i++)  
    {  
        int found = 0;  
  
        for(j=0;j<frames;j++)  
        {  
            if(frame[j] == pages[i])  
            {  
                found = 1;  
                pageHit++;  
                break;  
            }  
        }  
  
        if(found == 0)  
        {  
            int pos = -1;  
            int farthest = i;  
  
            for(j=0;j<frames;j++)  
            {  
                int k;  
                for(k=i+1;k<12;k++)
```

```
tanuja@Tanuja: ~  
GNU nano 6.2  
  
    for(j=0; j<frames; j++)  
    {  
        int k;  
        for(k=i+1; k<12; k++)  
        {  
            if(frame[j] == pages[k])  
                break;  
        }  
  
        if(k > farthest)  
        {  
            farthest = k;  
            pos = j;  
        }  
    }  
  
    if(pos == -1)  
        pos = 0;  
  
    frame[pos] = pages[i];  
    pageFault++;  
}  
}  
  
printf("Total Page Hits = %d\n", pageHit);  
printf("Total Page Faults = %d\n", pageFault);  
  
float hitPer = (pageHit/12.0)*100;  
float faultPer = (pageFault/12.0)*100;  
  
printf("Page Hit Percentage = %.2f%%\n", hitPer);  
printf("Page Fault Percentage = %.2f%%\n", faultPer);  
  
return 0;  
}
```

Output:

```
tanuja@Tanuja: ~  
tanuja@Tanuja:~$ nano optimal.c  
tanuja@Tanuja:~$ gcc optimal.c  
tanuja@Tanuja:~$ ./a.out  
Total Page Hits = 5  
Total Page Faults = 7  
Page Hit Percentage = 41.67%  
Page Fault Percentage = 58.33%  
tanuja@Tanuja:~$ |
```